

The structure of $\text{SiO}_2\text{--GeO}_2$ glasses: A spectroscopic study

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Abstract

Four glasses of the $\text{SiO}_2\text{--GeO}_2$ binary system have been synthesized via a sol–gel route followed by a heat treatment and a quench. Glass structure has been determined by Ge K-edge X-ray absorption spectroscopy (XAS) at low temperature and Raman spectroscopy. These mixed glasses present a continuous random network of interconnected GeO_4 and SiO_4 tetrahedra, with GeO_4 tetrahedra similar to the GeO_4 units in GeO_2 glass and continuous compositional variations from GeO_2 -rich regions to SiO_2 -rich regions. Such a random mixture is consistent with physical properties of these binary glasses as well as with the chemical dependence of their polyamorphism at high pressure. This EXAFS-derived mean Ge–O–Si angles are close to the Ge–O–Ge mean angle in GeO_2 glass, 134° and 130° , respectively. This misfit with the Si–O–Si angles might explain the ease of formation of isolated and pair defects centers, which are suspected to be at the origin of photo-induced modifications of optical properties in Ge-bearing SiO_2 glasses.

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1. Introduction

Germanosilicate glasses are used in a considerable range of applications in photonics, ranging from standard optical fibers for telecommunications to Bragg gratings or nonlinear optical devices [1]. The enhancement of defect production in GeO_2 -doped silica glass results in transparency loss and has led to the discovery of glass photosensitivity induced by UV exposure [2,3]. Vitreous germanates have also been proposed as materials for optical waveguides, due to their transmission characteristics in the infrared region. In addition, the high Raman scattering cross-section makes these glasses good candidates for devices such as fiber optic amplifiers and oscillators by employing stimulated Raman scattering. Though the presence of Ge atoms is suspected to modify the atomic structure of silica and to cause

both permanent and photo-induced variation of the refractive index, the microscopic origin of the original properties of germanosilicate optical fibers and glasses is still poorly constrained. An earlier EXAFS study of silica-rich compositions [4] has shown a local structure based on a continuous network of interconnected SiO_4 and GeO_4 tetrahedra. This model has been confirmed by molecular dynamics [5], although high energy X-ray diffraction measurements on a mixed glass with $\text{Si/Ge} = 2.4$ question the continuous nature of this random network [6]. The extent of the Si/Ge mixing and the properties of the T–O–T distribution (T = Si or Ge) are still poorly known, despite the importance of these parameters in the control of the formation of photo-induced defects. These oxygen-deficient defects, at the origin of the innovative properties of these materials, are often associated to Ge, e.g. the Ge E'-center or the divalent Ge defect. An additional difficulty raised by these glasses is their chemical homogeneity due to the difficulty of producing glasses by a mixing of highly viscous, poorly reactive melts.

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This paper describes the preparation protocol and structural results obtained on four glass compositions over the SiO_2 – GeO_2 binary. These glasses have been synthesized from a sol–gel route followed by a heat treatment and a quench, and investigated using Raman spectroscopy together with Ge K-edge X-ray absorption spectroscopy (XAS) at low temperature to get rid of thermal disorder effects. These data confirm the adequacy of a continuous random network (CRN) model for these glasses, with a homogeneous Si/Ge distribution. The Ge–O–Si bond values illustrate the constraints arising from the mixing of two topologically different 3-dimensional networks. The existence of a random mixture is consistent with the physical properties of these binary glasses as well as with the chemical dependence of their polyamorphism at high pressure. The local misfits resulting from the accommodation of Ge in Ge-bearing silica glasses may explain the disappearance of the ring defects characteristic of the structure of pure silica. They constitute also a possible precursor for the formation of photo-induced structural defects.

2. Experimental

2.1. Synthesis of the SiO_2 – GeO_2 glasses

$(\text{SiO}_2)_x(\text{GeO}_2)_{100-x}$ glasses, x representing the amount of SiO_2 , have been prepared using a two-step synthesis procedure with $x = 30\%$, 50% , 65% and 80% . The first step involved a sol–gel technique using tetraethoxysilane $\text{Si}(\text{OC}_2\text{H}_5)_4$ (TEOS) and tetraethoxygermane $\text{Ge}(\text{OC}_2\text{H}_5)_4$ (TEOG) as precursors of the SiO_2 and GeO_2 components, which is close to that described in Ref. [1] for coating deposition purpose. The appropriate amount of TEOS was first diluted in anhydrous ethanol, and was pre-hydrolyzed with two molar equivalents of H_2O at pH 2, during 2 h at room temperature. Then, the prescribed amount of non-diluted TEOG was added dropwise, and the solution was stirred during 30 min. After this time, a post-addition of water was done in order to remove the residual M–OR groups and extend the oxo-based network. The total amount of water (pH 7) was however still sub-stoichiometric with $\text{H}_2\text{O}/\text{M} = 2$, because of the high reactivity of TEOG toward hydrolysis. Gelation occurred in a time interval of 5 min up to one day for the solution of highest TEOS content. The gels were allowed to age in an ethanol atmosphere at room temperature for 2 days, and left in air for about one month, during which they dried and shrank up to more than 50 vol.%. The dried gels with $x = 30$ and $x = 50$ were partially crystallized, while the other two were determined amorphous using X-ray diffraction analysis. The second step of the procedure aims to dry thoroughly the gels and remove the pores generated by retention of the solvent. For this purpose, the gels were heat-treated at high temperature between 1400 and 1520 °C during 40 min, and the materials were quenched by immersion of the bottom of the crucible in water. The morphology of the resulting materials indicated a shape modification of the gels above

the glass transition. The resulting glasses were amorphous by X-ray diffraction analysis. The chemical composition of the synthesized glasses has been determined by microprobe analysis (Table 1) and did not deviate from the nominal one by more than 2 mol%. In the following, the different glasses are designated SG x , with x representing the amount of SiO_2 (in mol%).

2.2. XAS and Raman spectroscopies

Ge K-edge XAS spectra for the SG glasses and GeO_2 quartz (q- GeO_2) have been recorded on the D44 beamline of the Laboratoire pour l'Utilisation du Rayonnement Electromagnetique (LURE, Orsay, France), equipped with a channel-cut Si(111) monochromator, and used in a transmission mode with ionization chambers for I0 and I1 measurements. Uniform layers of powdered samples were prepared by microporous filtering from an ethanol solution. During the measurements, a helium cryostat cooled the samples down to ~ 12 K, in order to minimize the thermal disorder. The extended X-ray absorption fine structure (EXAFS) signal (χ) was first extracted using a Victoreen function and a spline function to model the pre-edge and atomic absorption background signals, respectively. The $k^3(k)$ signal was Fourier transformed on the interval 4–17.7 Å^{−1} with a Kaiser window ($\tau = 2.5$). Inverse Fourier transforms were then performed to filter the EXAFS signal corresponding to the first peak (Ge–O, interval 0.8–1.8 Å^{−1}), the second peak (Ge–second neighbors, interval 2.2–3.5 Å^{−1}) or both (interval 0.8–3.5 Å^{−1}). Phase and amplitude parameters for the Ge–O shell as well as the Ge–Ge and Ge–Si shells were calculated with the FEFF6 program [7], on the basis of the quartz structures for GeO_2 and SiO_2 compounds (Ge being the absorbing atom substituting Si in q- SiO_2). Details for the EXAFS fitting are given below.

The Raman measurements were made with a T64000 Jobin–Yvon confocal microRaman spectrometer equipped with a CCD detector. The 514.532 nm line of a Coherent 70-C5 Ar⁺ laser operating at 2.5 W was used for sample excitation. For the samples studied here, the excitation and CCD system result in signal-to-noise ratio of 80/1. The integration time was 30 s. The triple spectrometer makes it possible to go down to low frequency (<10 cm^{−1}). All spectra are unpolarized. Spectra were corrected for the temperature and frequency-dependent scattering intensity before statistical analysis [8]. The corrected Raman intensities were normalized to the highest absolute intensity.

Table 1
Molar compositions of the SiO_2 – GeO_2 glasses determined by microprobe analysis (average of 10 analyses per sample)

	SG30	SG50	SG65	SG80
SiO_2	27.5	49.8	64.8	82.2
GeO_2	72.5	50.1	35.2	18.1

3. Results

3.1. Low temperature Ge K-edge XAS spectra

The Ge K-edge XANES spectra of GeO₂ quartz (q-GeO₂), GeO₂ glass (g-GeO₂) and SG30 glass are compared in Fig. 1. All SG glasses have a XANES spectrum identical to that of the SG30 glass. The white line and C features are similar in the three compounds in agreement with a tetrahedral coordination for Ge [9]. The A and B features present in q-GeO₂ are smeared out in g-GeO₂ and disappear in SG30 (Fig. 1). As they correspond to multiple scattering of the photoelectron beyond the tetrahedron, these features are characteristic of the medium-range ordering, which is less extended in SG glasses than in GeO₂ glass, a consequence of chemical and structural disorder induced by the Si/Ge mixing.

The Fourier-transform magnitude of the EXAFS signal obtained for the four investigated glasses show the contribution of the Ge–O coordination shell and Ge/Si second neighbors (Fig. 2). The Fourier-filtered EXAFS signals corresponding to each peak have been fitted at first separately, then together using three shells (Ge–O, Ge–Ge and Ge–Si) with a harmonic model [10]. This approach turned out to be relevant for the Ge–O coordination shell, and good as a first approximation for the second Ge–T shell, as demonstrated by the fits in Fig. 3.

The Ge–O distance ($d_{\text{Ge-O}}$), coordination number (N) and Debye–Waller factors (σ), determined from the filtered EXAFS signal of the first peak, are reported in Table 2. These structural parameters remain identical whatever the composition, and point out a slightly distorted GeO₄ tetrahedron, with a mean Ge–O bond length of 1.73 Å and a Debye–Waller factor of 0.051 Å at 12 K. The Ge–O distances are similar to those found in earlier studies [4,11].

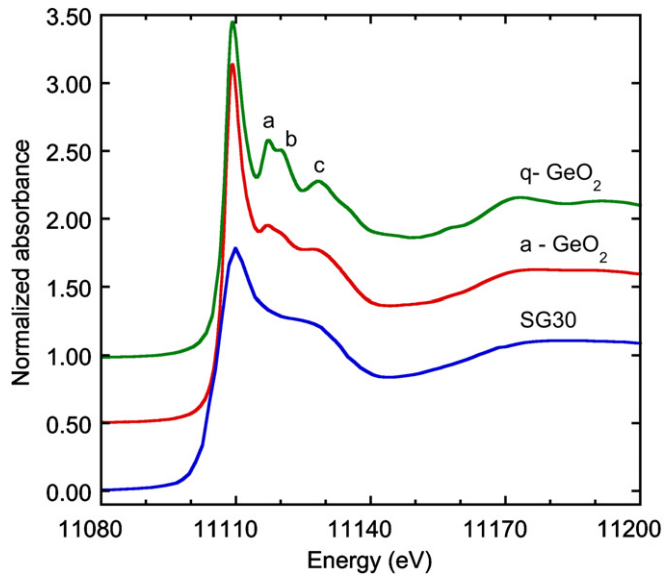


Fig. 1. XANES spectra of GeO₂ quartz (q-GeO₂, top), GeO₂ glass (a-GeO₂, middle), and the (SiO₂)₃₀(GeO₂)₇₀ glass (SG30, bottom).

The distribution of the distances (Ge–T, with T = Ge or Si) gives rise to weak EXAFS contributions. Fitting the second shell separately shows that the presence of a Ge–Si component, in addition to the Ge–Ge component,

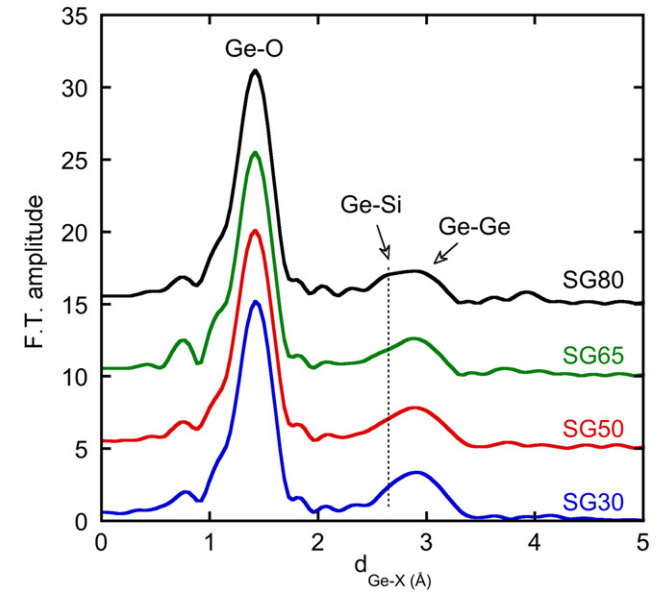


Fig. 2. Magnitudes of the Fourier Transforms of the EXAFS signals over the 4–17.7 Å^{−1} range.

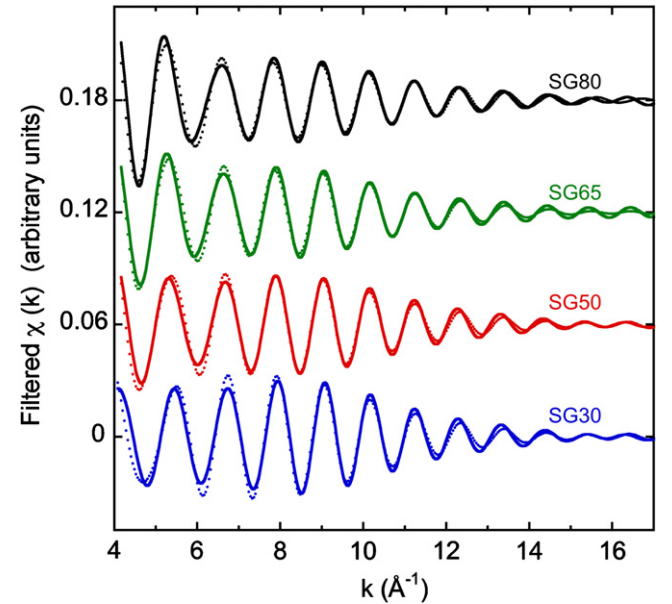


Fig. 3. Fourier-filtered EXAFS signals (lines) and 3-shells fits (dots).

Table 2 EXAFS parameters of the Ge–O coordination shell in the SiO ₂ –GeO ₂ glasses			
	$N \pm 0.4$	$\sigma \pm 0.001 \text{ \AA}$	$d_{\text{Ge-O}} \pm 0.005 \text{ \AA}$
SG30	4.3	0.052	1.730
SG50	4.2	0.053	1.730
SG65	4.2	0.051	1.730
SG80	4.4	0.051	1.730

results in a better fit at low k ($4\text{--}8\text{ \AA}^{-1}$). Indeed, the amplitude of the Ge–Si single scattering is high on this k -range, in contrast to the amplitude of the Ge–Ge one. Concerning the contribution of multiple scattering events, the Ge–O–O multiple scattering path, where both oxygen atoms belong to the same Ge tetrahedron, corresponds to a half distance of about $3.16\text{ \AA}$, in the same range as the Ge–T single scattering path. We have checked out that this path gives only a negligible contribution to the total XAFS spectrum, by modeling it in the GeO_2 quartz structure using FEFF6. In Table 3 are presented the parameters of the second shell obtained by fitting the three components together (Ge–O, Ge–Ge and Ge–Si), using as initial set of parameters, the values found in the separate fits of the Ge–O and Ge–T shells. The sum of the number of Si and Ge neighbors was fixed to 4, which is the value expected for a continuous random network of tetrahedra. The harmonic model is a first approximation but the following trends result from these fits: (i) the amplitude of the second shell signal can be well modeled assuming a continuous network of corner linked tetrahedra, with broad Ge–Ge and Ge–Si distributions having Debye–Waller factors of $0.06\text{--}0.1\text{ \AA}^2$ at 12 K ; (ii) the Ge–Ge and Ge–Si distances have the same mean values over all the compositions, i.e. $3.14 \pm 0.02\text{ \AA}$ and $3.07 \pm 0.02\text{ \AA}$, respectively and (iii) the number of Si neighbors increases continuously with increasing SiO_2 content.

3.2. Raman spectra

The normalized Raman spectra for all the investigated glasses are shown in Fig. 4. For 3D-glasses, the interpretation of the Raman bands usually refers to the vibration modes of the T–O–T inter-tetrahedral bonds. The spectra of the samples studied here are similar to previous data published on this system [12]. The symmetric stretching of the oxygen band shifts from 437 to about 415 cm^{-1} and progressively narrows as the GeO_2 content increases, which has been related to a decrease in the mean value and width of the T–O–T bond angle distribution. The region between 850 and 1250 cm^{-1} encompasses several bands assigned to the asymmetric stretching motion of the binding oxygen (ν_{as} (T–O–T)). This region contains information about the mixing of the SiO_4 and GeO_4 units, since the ν_{as} (T–O–T) bands involve the motion of T atoms

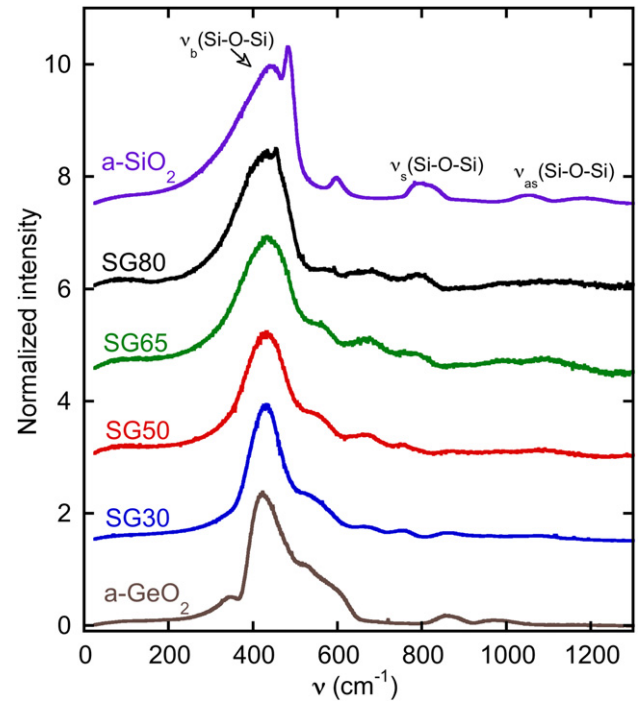


Fig. 4. Raman spectra of the $\text{SiO}_2\text{--GeO}_2$ glasses along with the a-GeO_2 and the a-SiO_2 spectra for comparison.

and thus their positions depend on the T mass (and on the T–O–T angle). Fig. 5 focuses on the $700\text{--}1300\text{ cm}^{-1}$ range. The ν_{as} (T–O–T) band for the pure glasses is divided into two components at 858 and 973 cm^{-1} (a-GeO_2) or 1060 and 1190 cm^{-1} (a-SiO_2), which arise from a TO–LO splitting of this mode. The LO mode causes the broad high-frequency part and is thought to be sensitive to the structural

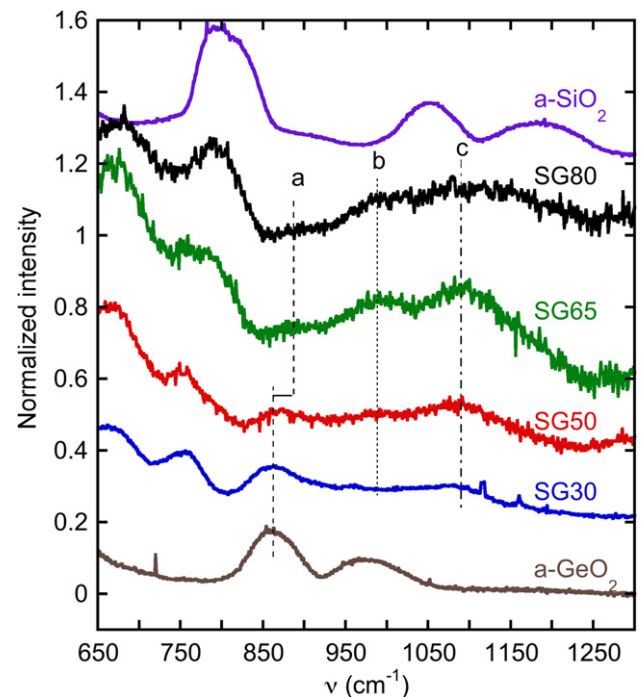


Fig. 5. Focus on the $800\text{--}1300\text{ cm}^{-1}$ region of the Raman spectra.

Table 3
EXAFS parameters of the Ge–Ge and Ge–Si shells determined from the 3-shell fitting shown in Figs. 2 and 3

	$N_{\text{Ge-T}} \pm 0.4$		$\sigma_{\text{Ge-T}} \pm 0.01\text{ \AA}$		$d_{\text{Ge-T}} \pm 0.02\text{ \AA}$	
	$N_{\text{Ge-Ge}}$	$N_{\text{Ge-Si}}$	$\sigma_{\text{Ge-Ge}}$	$\sigma_{\text{Ge-Si}}$	$d_{\text{Ge-Ge}}$	$d_{\text{Ge-Si}}$
SG30	2.55	1.45	0.099	0.057	3.14	3.07
SG50	1.9	2.1	0.098	0.063	3.13	3.07
SG65	1.2	2.8	0.081	0.080	3.13	3.07
SG80	0.8	3.2	0.066	0.098	3.15	3.08

$N_{\text{Ge-T}}$, $\sigma_{\text{Ge-T}}$, $d_{\text{Ge-T}}$ are the coordination number, Debye–Waller factor and distances corresponding to second neighbours T (T = Ge or Si) around Ge.

disorder of the glass [12]. In the mixed glasses, three successive bands are observed at about 880, 990 and 1100 cm^{-1} , which can be assigned to the ν_{as} (Ge–O–Ge), ν_{as} (Ge–O–Si) and ν_{as} (Si–O–Si) vibrations, respectively, on the basis of the atomic mass involved. The LO modes are assumed to be undetectable because of the high structural disorder introduced by the mixing. The ν_{as} (Ge–O–Ge) and ν_{as} (Si–O–Si) bands appear at higher frequencies with respect to the pure glasses, and the ν_{as} (Ge–O–Si) band very slightly shifts toward higher frequencies with the SiO_2 content. This may be caused by changes in the T–O–T distributions with the composition.

In the region between 520 and 800 cm^{-1} , we observe three main bands at about 570, 670 and 750–800 cm^{-1} . According to Ref. [12], the band at 570 cm^{-1} may stem from some Ge–O–Ge deformation mode, analogous to that giving rise to the shoulder at 558 cm^{-1} in GeO_2 glass [13]. Then, the broad band at 670–675 cm^{-1} is assigned to Si–O–Ge deformation modes. At last, the third band in our spectra shifts in position from about 750 cm^{-1} up to 800 cm^{-1} and becomes more intense as the SiO_2 content increases. We therefore assign it to the deformation modes of the Si–O–Si bonds, by analogy with the 800–830 cm^{-1} envelope in the SiO_2 glass. This band was not reported in the previous studies for glass compositions below 50 mol% SiO_2 [12,13].

4. Discussion

Both Raman and XAS data collected on SiO_2 – GeO_2 glasses are consistent with the model of a random network of corner-linked GeO_4 and SiO_4 tetrahedra, as described in pure SiO_2 and GeO_2 glasses [14,15]. There is little doubt in the literature about this issue, though it had been recently questioned in a high-energy X-ray diffraction investigation of a SiO_2 – GeO_2 glass (close to the SG65 composition), which concluded to an incomplete continuous random network [6]. The Debye–Waller factor for the coordination shell is similar in SiO_2 – GeO_2 glasses and in q- GeO_2 (Table 2). Similar Debye–Waller values have also been reported for the Ge–O pair in GeO_2 glass and in q- GeO_2 [16]. The static component of the Debye–Waller factor is close to that in q- GeO_2 , whose four Ge–O distances in the tetrahedron have a variance of 0.005 Å. Therefore, the distortion of the GeO_4 tetrahedron is similar in the mixed glasses and in the pure GeO_2 glass, by contrast to previous findings [6].

Raman and XAS spectra put in evidence the existence of Ge–O–Si bonds in SiO_2 – GeO_2 glasses. According to a random mixing model, the Ge–O–Ge, Ge–O–Si and Si–O–Si contributions vary as $2(1-x)$, $2x(1-x)$ and $2x$, respectively, in $x\text{SiO}_2$ – $(1-x)\text{GeO}_2$ glasses. The chemical dependence of the EXAFS contribution of the second neighbor shell is compatible with this model (Table 3). However, $N_{\text{Ge-Ge}}$, $N_{\text{Ge-Si}}$, $\sigma_{\text{Ge-Ge}}$ and $\sigma_{\text{Ge-Si}}$ parameters in Table 3 form a non-unique set of parameters for the same fit quality, and these EXAFS data cannot be conclusive. Besides, the observation that the ν_{as} (Ge–O–Ge) and ν_{as} (Si–O–Si)

bands are more intense than the ν_{as} (Ge–O–Si) band in the Raman spectrum of SG50 glass favors a picture in which GeO_2 -rich regions exist with SiO_2 -rich ones, *i.e.* there are composition fluctuations over the network. However, an inhomogeneous glass structure is unlikely on the basis of the general evolution of the symmetric stretching oxygen band (ν_{ss} (T–O–T)), which suggests a regular compositional dependence for the T–O–T distribution. The coexistence of two regions of different compositions should result in a broader ν_{as} (T–O–T) band. This is in agreement with a molecular dynamics simulation of a SG50 glass indicating a homogeneous distribution of the SiO_4 and GeO_4 units [5]. Such a random mixture is consistent with the linear variation as a function of glass composition of the physical properties of these binary glasses, including density, refractive index of thermal expansion coefficients [17]. It also explains the chemical dependence of the polyamorphism of GeO_2 – SiO_2 glasses at high pressure, which shows that structural changes take place at higher pressures than in pure GeO_2 glass, with increasing silica content [18,19].

Because of the distinct backscattering amplitudes for Ge and for Si, reliable estimates of Ge–Si and Ge–Ge distances are $d_{\text{Ge-Ge}} = 3.14 \pm 0.02$ Å and $d_{\text{Ge-Si}} = 3.07 \pm 0.02$ Å for the four samples, consistent with the mean T–T distance of 3.09 Å found in X-ray diffraction [6]. These distances imply a mean T–O–T bond angle of $130.3^\circ \pm 1^\circ$ and $133.6^\circ \pm 1^\circ$ for Ge–O–Ge and Ge–O–Si, respectively. These values are in close agreement with EXAFS parameters in glassy GeO_2 $d_{\text{Ge-Ge}} = 3.14$ Å and mean Ge–O–Ge angle = 130° [15,16]. A non-Gaussian distance distribution may explain the slight difference with the Ge–Ge distances determined by AWAXS in glassy GeO_2 (3.16 ± 0.03 Å and 132° [20]). The Ge–O–Si mean angle of 133.6° is close to the Ge–O–Ge mean angle in a- GeO_2 (130° using Ge K-edge EXAFS), but lower than the wider Si–O–Si angle in a- SiO_2 (148° according to Ref. [21]). Both Ge–O–Ge and Ge–O–Si mean angles indicate a medium-range environment for Ge in the mixed glasses close to that in pure GeO_2 glass.

The two Raman bands in SiO_2 glass, at 490 and 606 cm^{-1} , often referred to as D1 and D2, respectively, almost vanish with the addition of 20 mol% GeO_2 . These two bands have raised much effort but are now related to minor planar four members and three members silicate rings, respectively [22,23]. A Raman study performed on SiO_2 – GeO_2 glasses at low GeO_2 content revealed that the added Ge atoms preferentially insert within these defects [24], and destabilize them. The small values of the Ge–O–Si angle determined by EXAFS favor such a behavior. In GeO_2 -rich glasses, the narrow bands at 344 cm^{-1} and 520 cm^{-1} may arise from analogous rings [25], and also disappear with the addition of 30 mol% SiO_2 . The high potential and configurational energy localized on these small rings defects is converted into configurational energy by insertion of the foreign atoms, as they add degrees of freedom in the network. As a result, the mixed glasses contain fewer small-size rings than pure glasses.

The EXAFS structural parameters for the GeO_4 tetrahedron and the Ge–O distances do not vary significantly with the composition. This suggests that either the Ge local environment is independent on the composition over the SiO_2 – GeO_2 binary, or that Ge exists in regions of similar composition within an inhomogeneous structure. This latter hypothesis is not supported by the Raman spectra discussed above. The Ge cations may locally constrain the network to a more compact order, consistently with the greater ionic character of the Ge–O bond. The local constraints resulting from the misfit between Si–O–Si and Ge–O–Si angles might explain the ease of formation of isolated and pair defect centers associated to Ge in SiO_2 – GeO_2 glasses. These oxygen-deficient defects are suspected to be at the origin of photo-induced modifications of optical properties [26,27].

5. Conclusion

Four glasses in the SiO_2 – GeO_2 binary system have been prepared via a sol–gel route followed by a high temperature heat-treatment to drive the gels into the undercooled liquid state. The extent of the Si/Ge mixing can be assessed on the basis of the Raman spectra and is consistent with continuous variations of composition from GeO_2 -rich to SiO_2 -rich regions. The overall structure can be described as a continuous random network of GeO_4 and SiO_4 tetrahedra, with GeO_4 tetrahedra similar to those forming the pure GeO_2 glass network. The structural effect of the mixing affects the medium range order, with the presence of small Ge–O–Si angle values of about 134° . As a consequence of this disorder, the D1 and D2 bands disappear in the mixed glasses. The Ge local environment up to the first shell of neighbouring tetrahedra appears close to the Ge environment in a- GeO_2 . This misfit with the Si–O–Si angles might explain the ease of formation of isolated and pair defects centers in Ge-bearing SiO_2 glasses. These point defects are suspected to be at the origin of the appearance of technologically important photo-induced modifications of optical properties.

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References

- [1] D.G. Chen, B.G. Potter, J.H. Simmons, *J. Non-Cryst. Solids* 178 (1994) 135.
- [2] K.O. Hill, Y. Fujii, D.C. Johnson, B.S. Kawasaki, *Appl. Phys. Lett.* 32 (1978) 647.
- [3] B. Poumellec, P. Guenot, I. Riant, P. Sansonetti, P. Niay, P. Bernage, J.F. Bayon, *Opt. Mater.* 4 (1995) 441.
- [4] R.B. Gregor, F.W. Lytle, J. Kortright, A. Fischercolbrie, *J. Non-Cryst. Solids* 89 (1987) 311.
- [5] C. Bernard, S. Chaussedent, A. Monteil, N. Balu, J. Obriot, C. Duverger, M. Ferrari, M. Bouazaoui, C. Kinowski, S. Turrell, *J. Non-Cryst. Solids* 284 (2001) 68.
- [6] H. Schlenz, J. Neufeind, S. Rings, *J. Phys.: Condens. Matter* 15 (2003) 4919.
- [7] J.J. Rehr, J. Mustre de Leon, S.I. Zabinsky, R.C. Albers, *J. Am. Chem. Soc.* 113 (1991) 5135.
- [8] D.R. Neuville, B.O. Mysen, *Geochim. Cosmochim. Acta* 66 (1996) 1727.
- [9] J.P. Itié, A. Polian, G. Calas, J. Petiau, A. Fontaine, E. Tolentino, *Phys. Rev. Lett.* 63 (1989) 398.
- [10] K. Teo, *EXAFS: Basic Principles and Data Analysis*, Springer Verlag, Berlin, 1986, p. 349.
- [11] C. Lapeyre, J. Petiau, G. Calas, F. Gauthier, J. Gombert, *Bull. Minéral.* 106 (1983) 77.
- [12] S.P. Mukherjee, S.K. Sharma, *J. Am. Ceram. Soc.* 69 (1986) 806.
- [13] S.K. Sharma, D.W. Matson, J.A. Philpotts, T.L. Roush, *J. Non-Cryst. Solids* 68 (1984) 99.
- [14] G.N. Greaves, S. Sen, *Adv. Phys.* 56 (2007) 1.
- [15] M. Micoulaut, L. Cormier, G.S. Henderson, *J. Phys.: Condens. Matter* 18 (2006) R753.
- [16] M. Okuno, C.D. Yin, H. Morikawa, F. Marumo, *J. Non-Cryst. Solids* 87 (1986) 312.
- [17] E.M. Birtch, J.E. Shelby, *Phys. Chem. Glasses* 47 (2006) 182.
- [18] G. Calas, G.E. Brown, F. Farges, L. Galois, J.P. Itié, A. Polian, *Nucl. Instr. Methods B* 97 (1995) 155.
- [19] O. Majerus, L. Cormier, J.-P. Itié, L. Galois, D.R. Neuville, G. Calas, *J. Non-Cryst. Solids* 345 (2004) 34.
- [20] D.L. Price, M.L. Saboungi, A.C. Barnes, *Phys. Rev. Lett.* 81 (1998) 3207.
- [21] J. Neufeind, K.D. Liss, *Ber. Bunsengesellschaft* 100 (1996) 1341.
- [22] A. Pasquarello, R. Car, *Phys. Rev. Lett.* 80 (1998) 5145.
- [23] A. Rahmani, M. Benoit, C. Benoit, *Phys. Rev. B* 68 (2003) 184202.
- [24] X. Nian, X. Zhisan, T. Decheng, *J. Phys.: Condens. Matter* 1 (1989) 6343.
- [25] R.A. Barrio, F.L. Galeener, E. Martinez, R.J. Elliott, *Phys. Rev. B* 48 (1993) 15672.
- [26] J. Nishii, K. Kintaka, H. Hosono, H. Kawazoe, M. Kato, K. Muta, *Phys. Rev. B* 60 (1999) 7166.
- [27] T. Uchino, M. Takahashi, T. Yoko, *Phys. Rev. Lett.* 84 (2000) 1475.